

# EXAM ESSAY

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Critical  
Reflection on  
Website

# Introduction

The development of a Pokémon themed website was an opportunity for me to engage with a global digital culture. Taking what I knew about the Pokémon world and exploring it using D3.js libraries and other web development tools (HTML, CSS, JavaScript), my aim was to create an information hub for users to go and gain access to an assortment of information about their favourite Pokémon. I aimed to combine this with a slew of UI/UX design techniques to bring home that brand identity while being unique. I have always had a connection with the Pokémon brand, so I wanted to show that through the technical and artistic choices I made. This essay critically analyses those technical and design decisions made throughout the development process, while exploring their broader implications.

Before diving into the critical analysis, I would like to take a moment to assert my position on whether I succeeded or failed in certain aspects of this project. In terms of using the API (pokeapi.co) to develop my data visuals, I believe I created 3 unique visuals which are all connected in different forms which I will explore further on in this essay. In terms of technical capabilities within the website it has real-time verification in form fields, a dynamically generated navigation bar and footer layout and it contains consistent load times and optimized pages filled with content. However, due to time constraints there are features missing within the code and the website such as ARIA labels and consistent responsiveness, accessibility options are lacking and meta-tags are not prevalent, introducing the main of failure on my part being, lack of attention to smaller detail.

## Global Economics and Artistic Implications of Web Technologies and Design Techniques

In the modern world, where access to the internet is almost considered a human right, it becomes apparent that access to international audiences is more than just the exchange of information. It becomes a practice that transcends geographical and cultural barriers, allowing people to discuss ideas, argue positions, and most importantly, in this context, impact many people in just a few seconds. One of the most important things I have learned using collaborative tools such as GitHub and design websites such as Dribbble.com and Awwwards.com is that the sharing of ideas enhances the quality of those around them. By allowing people to share their ideas and having others view them, it's almost certain someone will provide criticism or feedback, and that accelerates innovation, creating an environment that promotes collaboration and fosters an inherent ability to improve the work of developers.

Something important to note is that, even though many ideas nowadays are created with the hope of financial gain, the importance of open-source platforms and tools

cannot be overlooked. These platforms allow for a level of economic efficiency that was previously unattainable for independent developers and designers. Open-source tools (D3.js, HTML5, etc.) over paid programs (WIX, WordPress) not only allow anyone to contribute to the online content but also significantly reduce the costs associated with building and maintaining high-quality websites.

This not only affects technical development but also the artistic aspects of website design. In terms of the design techniques I have used, my Pokémon-themed website is instantly accessible globally. I've made use of web technologies to create a user experience that is engaging while also presenting it in a universally understood format. Colour schemes, interactive visuals, and other features are laid out in such a way that any user can interact with them, without any previous knowledge of Pokémon. It is accessible even to users who are colourblind, allowing them to engage with the site in terms of interactivity.

While keeping in mind that many of my design choices are influenced by global standards, such as the navigation bar being at the top of the page (though I would have preferred a side panel for easier navigation across the website), web technologies such as CSS and JavaScript have enabled me to balance artistic expression with functional usability. To encompass my point and my design techniques were the following, an intuitive website which is appealing in terms of aesthetic while also ensuring ease of use and smoothness across the board.

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## UI/UX Design: Creating Meaning and Engaging Users

Interactivity serves as a core element of the platform and the user experience. When creating something meaningful while also keeping redundancy and consistency in mind.

### UI/UX Design Philosophy

My design philosophy as I have mentioned is creating a website that focuses on an intuitive and engaging user experience. One of the main things I like to focus on is making the user interface effortless, from the interactions to the content, I want the user to understand what they are dealing with as soon as they enter the site. The navigation bar, interactive buttons, and page structures are all positioned to be accessible at any point on the website, by making use of breadcrumbs, back to top buttons on longer pages and quick links. Colours are consistent and I try to present the content in a logical manner making use of icons and colours to guide the user. Interactivity is a big part of all my design choices, its always how can I make this part of the website more engaging or how can I make it affect their experience in a positive way, smooth animations and hover effects are easy to implement and are clear and concise in delivering instant communication.

## Design Choices and Their Meaning

One of the central design themes for the website is a calling card to the iconic Pokémon aesthetic, which helps create an emotional connection to the user. Pokémon imagery and symbols are in places where the audience can be exposed to them, such as the loading screen with the poke-ball or the random Pokémon characters around the website, which open a new door of engagement when people recognise the more popular characters (Pikachu, Bulbasaur etc..). The nostalgia references in the colour scheme and carefully selected imagery infused with a modern design allows for a deep immersion into the Pokémon universe but also reinforces the design continuity across the website. It's essentially a balance of tradition conspiring with novelty to create a sense of fresh comfort.

## Data Visualization Design

The website features three interactive d3.js visualizations, bubble charts, evolution trees and a scatter plot that all function by transforming live API data into visual formats. The use of colour, shape, interactive tool tips, filters are all inherent features to make complex data easy to interpret. For instance, the evolution tree uses the branching structure to mimic the in-game progress of evolving progressing in Pokémon games, this is rewarding for players who already know the game as they can view it from the lens of general understanding, while users who don't have a connection with the game gain a more knowledgeable relationship with the visual. Similarly, the type of distribution bubble chart can be filtered and manipulated which allows for a more dynamic data exploration. These data design choices are not only to clarify the data but also have the user actively involved with the discovery of the website and the connections they can make, just like in the Pokémon games themselves.

## The Role of Interactivity: Impact on User Experience and Data Interpretation

### Interactivity and User Engagement

As mentioned before, the interactive features on this website, such as clickable cards, toggle filters, and tooltips, are essential in keeping the user engaged. These features not only allow for a sense of control through the website's design but also add depth to an otherwise flat experience. While interactivity at its core promotes constant engagement, the challenge in ensuring all the elements are meaningful and intuitive can become overwhelming. In some cases, I ran into the problem of introducing one too many interactive options, which, during testing, became too overwhelming for users, potentially distracting them from the main purpose of the section. Striking the steady line of engagement and simplicity is an ongoing challenge. A future iteration that I think would benefit the website would be to streamline the interactive areas with a bit more static option.

### Enhancing Data Interpretation

Going back to the Data Visuals, these contain interactive elements that contribute to the user flow by allowing the data to be more flexible with what is shown. This design decision does come with some drawbacks. For example, the features may not feel intuitive enough, and key insights could be missed if users don't fully understand or engage with the visualizations correctly. Another issue, which I will explore more in the next section, was the responsiveness when viewing the data on smaller screens. Certain features do not come across the way they are meant to, compared to larger screens, such as tooltips that require a "hover/touch" to engage with. It becomes a trade-off between usability and interactivity, which, as mentioned before, static variations would solve, especially on mobile.

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## Challenges Faced During Development

One of the more pressing technical challenges I faced was responsiveness in terms of keeping the integrity of the website while working across multiple devices. While the layouts I had weren't complex, it didn't help that I had a lot of CSS to make the website readable, particularly with the visualizations, to get them to fit without breaking the visuals was incredibly tough. Additionally, integrating live data from PokeAPI wasn't difficult per se, but coming up with interesting ways to present the data was a challenge. The underlying complexity of learning D3.js while doing this added to the need to better my future development cycles to account for difficulties.

### Technical Challenge: Reducing Loading Times for Pokémon Type Data

One critical challenge encountered was the delay in loading Pokémon type data, primarily caused by the `fetchTypeDistribution` function. This function retrieves type information by making individual API calls for each Pokémon, which slows the process significantly when working with a large dataset of up to 1000 Pokémon. Since the data is fetched sequentially, the cumulative loading time becomes impractically long. To address this issue, I explored several strategies to optimize performance:

- **Parallel Fetching:** A major improvement was to fetch data in parallel rather than sequentially, allowing multiple requests to be processed simultaneously. This approach dramatically reduced the overall loading time and made the site more responsive.
- **Reducing Data Size:** Limiting the number of Pokémon retrieved can also reduce strain on the API and improve performance. By testing with a smaller subset of data, I could strike a balance between loading efficiency and data completeness.
- **Error Handling for Fetch Requests:** Adding error handling for each API call ensures that failed requests don't interrupt the entire data-fetching process. This approach enhanced reliability, allowing the site to load with available data even if some requests fail.

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## Alignment with Professional Practices in International Web/App Development

Aligning with industry standards can be a bit of a drag sometimes; however, using HTML5, CSS, and JavaScript helps to ensure compatibility across all browsers, which is essential for international audiences. However, it must be noted that meeting these standards is only one part of the overall picture. The second is that within that compatibility, the experience needs to be smooth. This was a challenge as some animations behave differently across different browsers. These discrepancies were big enough to be major issues, and without extensive testing on each platform, could be hard to resolve over a couple of weeks. A step I would take in the future to better meet international web development standards would be to prioritize these compatibility issues as a part of my responsiveness campaign across the website.

### International Web Trends

The project aligns with several modern trends, including JavaScript frameworks (D3.js libraries) and interactive data visualizations. Yet, while incorporating these trends makes the site feel current, it also presents some challenges. For example, using D3.js for visualizations required significant customization, and the heavy reliance on JavaScript affected load times on lower-end devices, which can be an issue for users in regions with limited internet speeds. This experience suggests that while adopting popular trends can enhance user engagement, balancing these trends with performance and accessibility considerations is essential for global reach.

This project is the first time I have worked within the modern JavaScript frameworks. These modern trends are being used to make sites feel current while also requiring a significant amount of invested time. D3.js libraries and visuals can be tough to work with, but Andre mentioned that you are only limited by creativity; however, I would like to mention that you are also limited by time (haha). On a serious note, the experience that I gathered from adopting popular trends in JavaScript libraries on how to enhance user engagement was essential in me establishing an understanding of global standards.

### Collaborative and Agile Practices

Working independently limited the amount of feedback I could gather, and at times, it led to overlooking usability issues that could have been identified in a collaborative setting. This experience highlighted the importance of collaboration, even in individual projects, and suggested that in future work, seeking feedback from peers or beta testers could significantly improve the development process.

Working independently truly limited the amount of testing I could do. Within my scope of web development and web users, I could only see my website from my point of view,

even when trying to fit into the other person's shoes. Usability issues were diagnosed when I got people to test the website; however, the significance of global standards, where feedback loops and beta testers or peer testing could have significantly improved my development process.

## Conclusion

In conclusion, when understanding web design as a whole, the sub-categories of its personality can be broken down and observed as I have done above. This reflection was a thorough and complete understanding of what the process was like to build this website; however, it must be noted that I did have fun. At some point, it turned from work into a passionate academic project, which I pursued with the greatest of interests. It helps that I recently got back into Pokémon, so the passion was driven by a solid understanding of the world and its personality. To say I faced a few challenges is an understatement. Struggling with D3 libraries and JavaScript as a whole was something that affected my whole semester in terms of coding. But I brought my best foot forward on this in terms of my current skill level. In years to come, I might look back at this website and shudder (in terms of some of my questionable design choices) but for now this is my final submission.